

E-tivity 1



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Risk, Ethics and Governance of AI

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0.1 Task 2

The myth of the golem, marked in the infographic as **ancient**, has been part of Jewish folklore dating back to the Old Testament (Rozenberg and Leviant 2007). My favourite retelling of the story is Wisniewski (1996)’s:

“A saintly rabbi miraculously brings to life a clay giant who helps him watch over the Jews of sixteenth-century Prague.”

In my opinion, an historically overlooked attribute of the golem is its vision. In fiction, the ability to see and understand the world comes as expected; the golem is a guardian, its vision is required. As a computer engineer, over the last three years, I have been involved in a road management project and I can say from first-hand experience that incident detection is not an easy task. The radars can only take you so far and now more than ever, the idea of casting silicon-made giant golems that can watch over the roads sounds less like magic and more like another AI-powered solution.

Motorways have the highest speed limits and cover the longest distances. Have the AI vision models reached the trustworthiness of the fantastical golem’s eyes? Moreover, road management is governmental and political; would nations accept help from an automaton? Or would politics push the creators to undo their golems? just like the demise of Joseph the golem, terminated by his creator after a political treaty (Wisniewski 1996). Before sharing my thoughts, I want to review the recent history around this topic.

In **2006**, the UK Parliament implemented the first trial on smart motorways on the M42. From junction 3a to junction 7 (UK Parliament 2019), inductive loop detectors and variable message signs (VMS) helped control traffic flow on the hard shoulder. A useful solution for relieving congestion, however, despite the UK Parliament calling it ‘smart’, to today’s standards, the term seems like a stretch. And no vision component was part of it. But when put into context, this same year, NVIDIA was releasing for the very first time a new parallel computing platform: Compute Unified Device Architecture, or CUDA (NVIDIA 2026).

CUDA would eventually become a complete game-changer; it will allow software developers to reach computational levels never even dreamed of before, but

not in 2006. However, another important event from this year is that Dr. Fei-Fei Li Deng, often referred as the ‘Godmother of AI’ (especially for her contributions to the computer vision discipline), was already formulating rough ideas of a new project she thought necessary.

In **2009**, ‘ImageNet: A large-scale hierarchical image database’ was formally introduced to the computer vision community (Deng et al. 2009). Dr Li and her colleagues knew that large amounts of data would be key to developing and testing vision models. And they were not wrong. Just a year later (**2010**), they were organising the first edition of ‘ImageNet Large Scale Visual Recognition Challenge’ or ILSVRC (Russakovsky et al. 2015). Initiatives like this were great for the machine learning community, bringing it focus and competitiveness. For everyone else, it probably did not mean much, but for this niche, the air became electric and something huge was bound to happen. It did not happen the next year (2011), but the year after.

In **2012**, AlexNet (Krizhevsky et al. 2017) won the ILSVRC classification task, beating the second place (the model SIFT + FV) with a margin of 11.1% less error. This shook the foundation of object classification models in a way never seen before. It was the perfect storm that had been forming over the vision horizon for the last six years. Better hardware (thanks to CUDA giving access to the GPU) and rescued ideas from the past (of mimicking the human brain) made possible nearly a 50% reduction in object detection error, something unheard of before the use of deep learning and convolutional neural networks (CNN).

In this was not a fluke either, the next year (**2013**), the winners of ILSVRC also made use of CNNs to an absolute drop of 3.6 percentage points, or about a 23.5% relative reduction in error in comparison to AlexNet (Zeiler 2012). As time passed, different combinations of deep learning techniques and hardware GPU coding allowed for a greater accuracy to creep in.

In **2015**, the entity Transport Infrastructure Ireland (TII) was established (Transport Infrastructure Ireland 2015). This was a merger of the Railway Procurement Agency. It brought some modernisation to the country. TII took historical data from its parents and also made it available to the public. Here, I believe it is noticeable how concepts of machine learning have permeated even the highest places in the government. Data is valuable, and everybody knows it.

In **2017**, transformers shifted interest towards attention (Vaswani et al. 2017). It was too soon to know it, but transformers would revolutionise machine learning once again. By **2020**, a new framework computer vision framework, DERT: End-to-End Object Detection with transformers (Carion et al. 2020), already showcases how powerful this model (originally a natural language processing model) really is. And in **2023**, ChatGPT: Generative Pre-trained Transformer (Stanford Institute for Human-Centered Artificial Intelligence 2024), swept the world off its feet and changed the world.

And politicians noticed it. In **2024**, with the AI Act, Europe declared smart motorways (among others) as high-risk systems:

AI systems intended to be used as safety components in the management and operation of critical digital infrastructure, road traffic, or in the supply of water, gas, heating or electricity. (European Parliament and Council of the European Union 2024, Annex III, point 2).

Does this mean that smart motorways are bound to stay as smart as the UK Parliament thought they were in 2006? No, I believe that the AI Act is just calling for reflection in a never-before-seen scenario. Technology is advancing so fast that it is scary, but high risk does not mean ‘unacceptable risk’. Websites like <https://traffic.tii.ie/> or <https://umferdin.is/> are examples of already good intelligent motorway systems. The latter is great for tourists visiting Iceland, it allows people to check the road conditions in the whole country, in a place where you do not want to be caught in the snow.

The data collection is already happening. Strong efforts are going into sensor fusion; soon, features like halted vehicle detection will be trivial, and maybe the causality reductions will outweigh the risks. The UK wants to wait until 2028 to have enough data (Office of Rail and Road 2026).

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